CARRT Assistive Arm CSE Project — Weekly Status Report #8

Current Report Period: Weeks #6-8

1. Goals for the Current Period?

Carry-Over Goals (unaccomplished from prior reporting periods)

1. **Gazebo ADL Simulation Stabilization, Week #5** – [✓ | ✗] – Stabilize current code and achieve successful, repeatable simulation with a 30-50% success rate over multiple tests.
   - ***a. Reason for Carry-Over:*** The ADL execution was not completed by Week 5, due to time being spent rebuilding the motion planning package after Python packages were notably missing from ROS2 Jazzy software. This left little time remaining to do task scripting and testing.
   - ***b. Approach:*** With the environment and planning library now stable, the ADL task scripting may be further distributed over team members. The building of each ADL task will be completed using helper_moves, vision_stub, and scene testing infrastructure.

New Goals

2. **Design Review Presentation –** [✓ | ✗] – Prepare and deliver the Design Review presentation in the class, with a PowerPoint and rehearsed scripts for Week 6.
   - ***a. Approach:*** Divide slides by project component, and rehearse prior to delivery of presentation. Completed as expected.
2. **Tablet UI Wireframes –** [✓ | ✗] – Design a wireframe based off of previous design choices, to implement the ROS2 packages. Capture layout, interaction flow, and voice command triggers (button press).
   - ***a. Approach:*** Wireframe creation led by Brielle, based on previously decided strategy from last period. Shared drafts with team before finalization.
3. **Specification Document –** [✓ | ✗] – Complete and submit the Specification document as required, covering system design, constraints, trade-offs, risk analysis, and traceability to requirements and needs.
   - ***a. Approach:*** Divide sections by component ownership. Oliver leads architecture and design (Brielle for UI node), full team reviews and contributes before submission.

2. Goals Accomplished in the Period?

1. **Task #1 –** [Accomplished / **Partially**]: The core motion planning library was fully rebuilt using the MoveGroup action server interface, rather than the moveit_py file due to dependency issues in the environment. Coding components: apriltag_key, static_scene, vision_stub, and scene_from_vision were completed and verified in simulation. The clear_table task node is actively in development, and has successfully grasped only some of a set of objects, and so full repepatable execution of the tasks has yet to be achieved.
   - *a.* ***Completed By: Oliver (~8hrs), Jadon (~6hrs), Allison (~2hrs)***
2. **Task #2 –** [**Accomplished** / Partially]: The Design Review PowerPoint was completed and the presentation delivered in class during Week 6.
   - *a.* ***Completed By: Entire Team (~2hrs each)***
3. **Task #3** – [**Accomplished** / Partially]: The tablet UI wireframes were completed and distributed to the group members through GitHub. Brielle led wireframe design and development, collaborating with additional students with UI background to finalize layout and interaction flow. Also worked on button/control-mapping using voice and button commands.
   - *a.* ***Completed By: Brielle (~3 hrs)***
4. **Task #4** – [**Accomplished** / Partially]: The specification document was completed and submitted on March 6, 2026. The document was written to cover all specifics of our design, from entire design implementation to trade-offs and otherwise.
   - *a.* ***Completed By: Oliver (~3 hrs), Allison (~1hr), Jadon (~1hr), Brielle (~2 hrs)***

3. Critical Reflection on Unaccomplished Goals for the Period

1. **Task #1 –** Full execution was not achieved: The team underestimated the number of intermediate engineering steps that would be necessary to complete task execution. The MoveIt Python bindings library was not available on Jazzy, and we were forced to change to the MoveGroup action server method, rewriting a major block of code. We also needed scene planning infrastructure in order to test in the simulator.
   - ***a. Revised Approach:*** With infrastructure now working as intended, ADL scripting will be the sole focus over the next period. Tasks will be divided over group members. We will again target a 30-50% execution success rate before the next demo.

4. Major Goal List for Next Period (End of Period)

1. **ADL Task Execution –** Continue implementation and successfully test 2 of 3 ADL task scripts using Rviz and MoveIT. Target a repeatable execution success rate of 30-50%.
   - ***a. Approach:*** Explain plan of action / approach.
2. **UI Node Integration –** Begin integration of the UI node with the ROS2 tasks, verifying the task selection from the UI correctly triggers the ADL task execution.
   - ***a. Approach:*** Brielle connects the UI wireframe to the /adl_command topic, and test with a manual publish to confirm the task node receives and responds to commands.
3. **Test Plan Document –** Design and submit a test plan document to define key metrics and procedures on project testing.
   - ***a. Approach:*** Each team member works to draft the criteria for the ADLs and consolidate into one file.

5. Work Division for the Team (Over Entire Period)

- *Divisions are specifically noted in Sections 2 and 3. For each team member, by total hour count for the current week:*

Name:	Oliver Gaston	Jadon Orr	Allison Moline	Brielle Ashmeade
Hours Total:	~14 hrs	~9hrs	~5hrs	~7hrs